

1 Reliable comparison of severe felt-intensity
2 classifiers with versioned and dependent event
3 data: a USGS DYFI benchmark

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9 **Abstract**

10 Engineers comparing machine-learning classifiers on evolving event catalogues
11 can reach false conclusions when dependent events cross data splits or small score
12 differences are treated as wins. We ask whether candidate severe felt-intensity
13 classifiers comply with predefined rules for allowed inputs and evaluation, and
14 whether their apparent improvements remain supported after catalogue version,
15 event grouping, temporal evaluation, and paired uncertainty are fixed. Using a
16 frozen USGS “Did You Feel It?” (DYFI) snapshot of 7,520 $M \geq 5$ records, we
17 formed 2,362 eligible events, reserved 430 events from 2023–2024 as a tempo-
18 ral holdout, and evaluated fixed baselines using Brier score and sequence-cluster
19 bootstrap intervals. All four learned models improved on the no-skill reference;
20 paired uncertainty did not distinguish the two lowest-Brier models, and the input
21 rules excluded a candidate using event year. For a fixed random-forest diagnostic,
22 the observed differences between sequence-grouped and random folds were small
23 (Brier -0.006 ; area under the receiver-operating-characteristic curve [AUROC]
24 $+0.012$); this comparison is descriptive because no interval was retained. The
25 benchmark helps engineers reject inadmissible comparisons and unsupported win-
26 ner claims. It is intended for transparent retrospective comparison; confirmatory
27 or operational use would require fresh hidden data and separate validation.

28 **Keywords:** model comparison, evaluation reliability, benchmark qualification,
29 dependent-event data, reproducibility, USGS DYFI

1 Introduction

Engineering model selection depends on comparisons that remain valid when a dataset is updated, observations are dependent, and leading scores are close. Without those controls, an apparent improvement can be caused by a changed population, a proxy for the outcome or evaluation axis, cross-split dependence, or unquantified sampling variation rather than by a better model. Standard benchmarking is valuable precisely because it fixes a common task and metrics; recent work in this journal, for example, compares object detectors on COCO to support model selection for engineering applications [29]. The reliability problem is harder for event catalogues because both the catalogue and the dependence structure can evolve.

The U.S. Geological Survey “Did You Feel It?” (DYFI) system is a useful engineering case study. It converts crowdsourced felt reports into a community-decimal intensity (CDI) for each earthquake and is widely reused as an interpreted data product [2, 22, 34, 35]. The target studied here is retrospective classification of whether event-maximum CDI reaches 6 from outcome-blind source metadata. It is not a real-time alert, causal model, or operational triage rule.

Our engineering question is: *do performance and model-superiority claims remain supportable when catalogue version, dependent-event splitting, temporal evaluation, paired uncertainty, and reconstruction are controlled?* The intended users are applied machine-learning engineers evaluating a candidate classifier, maintainers publishing a stable benchmark, and reviewers auditing a claimed improvement. They need two decisions: whether a candidate is admissible for comparison and whether its improvement over a declared reference is supported rather than a point-score fluctuation. Existing felt-intensity studies use random or stratified partitions and do not provide that reusable decision contract [15, 20].

1.1 Contributions

We address unreliable model comparison by releasing a versioned benchmark and an executable candidate-qualification contract. Concretely, the work provides:

1. **A single-source, version-frozen DYFI benchmark** for the task “outcome-blind source metadata $\rightarrow$ later event-maximum $\max\text{CDI} \geq 6$ ”, frozen to one dated access instant so that any later catalogue revision defines a new version rather than a silent overwrite (Section 3).
2. **An executable six-category leakage audit** that rejects direct-label, post-outcome-ascertainment, identifier-proxy, and downstream-product features, and rejects cross-role sequence straddles and temporal-cutoff violations, accepting *only* the exact source-only feature set; it is exercised by 110 unit tests and rejects every deliberately contaminated test case (Section 4.1).
3. **Internally specified, outcome-independent roles** built on order-stable space-time sequence grouping and a temporal external holdout, so that split membership never uses the label (Section 4.2).
4. **A fixed, non-selective baseline suite** with calibration-aware proper scoring, Brier-score sequence-cluster bootstrap uncertainty, explicit handling of ties and unavailable calibration estimates, and single-use authorization of the decision-bearing protected evaluation (Sections 4.4–4.5).
5. **A candidate-qualification workflow** that fails closed on forbidden features or invalid roles and supports an improvement only when paired Brier-score uncertainty excludes zero (Section 5.5).
6. **A quantified split-sensitivity diagnostic** (naive random versus sequence-grouped splitting), together with a reproducible release bundle and machine-readable verification record (Section 5).

The learned models are worked candidates, not proposed production systems. Their comparison demonstrates the decisions the contract permits: all admissible

Table 1 Engineering failure modes addressed by the benchmark qualification contract

Failure source	Invalid engineering conclusion	Qualification control
Catalogue revision	Non-identical event population	Frozen access instant + SHA-256 release
Outcome-derived field	Label leakage	F1/F2 schema audit
Identifier or downstream proxy	Proxy leakage	F3/F4 field audit
Sequence crosses roles	Dependent-event contamination	F5 sequence-role audit
Temporal-cutoff violation	Future information in development	F6 temporal audit
Point-score ranking	False model-superiority claim	Paired sequence-cluster interval
Environment drift	Irreproducible scores/artifacts	Pinned environment + reconstruction hash

82 learned candidates improve over no-skill, one candidate is outside the declared
83 source-only schema, and the two lowest-Brier models remain unresolved.

84 1.2 Engineering significance and intended use

85 The output is an engineering test harness for transparent retrospective model compar-
86 ison. A declared candidate enters through the released schema, is excluded if it uses
87 forbidden outcome-derived fields or violates role isolation, is scored on the fixed tempo-
88 ral holdout, and is compared with paired sequence-cluster uncertainty. This workflow
89 prevents an engineer or reviewer from declaring superiority from a different catalogue
90 version, a contaminated split, or an unresolved point difference. Because holdout labels
91 are public, the release is not an untouched prospective test: confirmatory evaluation of
92 a new model requires a fresh benchmark version or hidden evaluation service. Table 1
93 names the failure prevented by each control, and Fig. 1 shows the decision path.

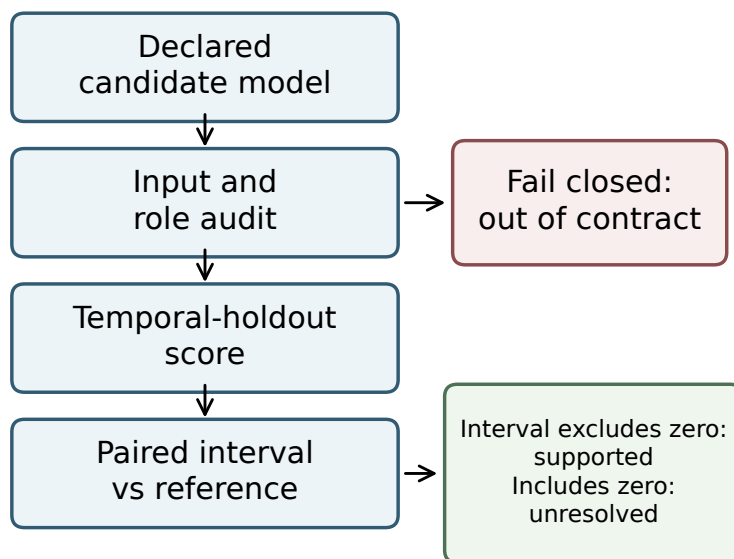


Fig. 1 Candidate-model qualification. A candidate that fails the schema, proxy, role, or temporal audit is inadmissible. An admitted candidate is scored on the fixed holdout and compared with paired uncertainty; an interval including zero remains unresolved rather than becoming a winner claim

2 Related work and novelty

DYFI and reusable macroseismic data.

Foundational DYFI work defines community decimal intensity, quality control, response dependence, and post-event revision [2, 22, 34, 35]. Rösler et al. [24] therefore recommend preserving timestamped interpreted products. Related resources include INGe and the Western North America macroseismic release [6, 18], historical databases [16, 27], and crowdsourced analyses of shaking and geology [12, 25]. These works motivate a frozen, reusable evaluation resource but do not define the present event-level classification contract.

Machine-learning and engineering comparisons.

Felt-intensity models have used random forests or boosting with instrumental, catalogue, or crowdsourced predictors [5, 14, 15, 19, 20]. The closest same-source study

106 uses a naive split for DYFI regression [10]; a concurrent DYFI benchmark studies
107 per-location MMI simulation and identifier leakage [13]. EarthquakeNPP supplies
108 the nearest broader leakage-aware seismological benchmark but targets occurrence
109 forecasting [28]. JEAS precedents demonstrate decision-oriented benchmark and
110 model comparisons in object detection, rock properties, and pavement maintenance
111 [1, 17, 29], while seismic engineering work makes the affected response and design
112 interpretation explicit [26]. The present study adopts that decision-oriented form for
113 model-comparison reliability rather than proposing a new classifier.

114 *Evaluation-method delta.*

115 Grouped or blocked validation for dependent data and leakage prevention are estab-
116 lished principles [11, 23]. The delta is the released combination for this task: a
117 timestamped and hashed single-source corpus, executable feature/proxy and split
118 audits, outcome-independent sequence-grouped roles, a temporal holdout, fixed base-
119 lines, paired sequence-cluster uncertainty, and exact reconstruction. No individual
120 component is claimed as new; the contribution is the auditable engineering comparison
121 contract and its worked decision example.

122 **3 Data, provenance, and outcome-boundary** 123 **disclosure**

124 *Source and single-instant pinning.*

125 The corpus is derived entirely from USGS DYFI and its parent ANSS Comprehensive
126 Catalog (ComCat), retrieved through the International Federation of Digital Seis-
127 mograph Networks (FDSN) event web service [30, 33]. The acquisition envelope is
128 outcome-independent: $M \geq 5$, product type `dyfi`, event type `earthquake`, starttime
129 2015-01-01 to endtime 2025-01-01. FDSN defines the end bound as inclusive; no frozen
130 record occurs at that boundary, so the realized snapshot spans 2015–2024. Results are

ordered by time. The preliminary count request and the final data request use this same envelope. Because the catalogue evolves, the downloaded data were frozen at one access instant (2026-09-01T06:33:30Z) and assigned a SHA-256 digital fingerprint. A preliminary count changed from 7,575 to 7,576 during the acquisition session, documenting one live-catalogue revision. The final downloaded snapshot contained 7,520 DYFI records, and a count requested 50 seconds later in the same acquisition session also returned 7,520. Thus, the one-record change belongs only to the earlier preliminary checks; it is not a discrepancy in the frozen analysis snapshot. Likewise, the 56-record difference between the last preliminary count and the final 7,520-record snapshot is not an analysis exclusion: the preliminary count had no paired payload, so its membership difference cannot be decomposed and is retained only as catalogue-version provenance. The near-contemporaneous final query and count agree exactly. The digital fingerprint of the downloaded bytes, rather than a preliminary count, defines this resource version. The frozen snapshot contains 7,520 DYFI records (Table 2, Fig. 2).

Protocol chronology.

The file records an internal 2026-09-01 freeze but was first publicly timestamped on 2026-09-19. We therefore describe it as an internally frozen protocol, not an externally registered preregistration.

Retrospective eligibility and post-report ascertainment.

DYFI intensity is an after-the-event quantity whose reliability grows with the number of felt reports [22]. To keep the label meaningful we apply a data-informed eligibility filter of at least ten responses per event. The response count is used *only* for eligibility and is explicitly forbidden as a predictive feature, because it is a post-outcome ascertainment proxy; this forbiddance is enforced by the leakage audit (family F2, Section 4.1). Eligibility yields 2,362 events with severe prevalence 0.298 (703 severe events). Of these, 2,360 events are assigned to the internally specified analysis roles

(Section 4.2) and contain 702 severe events; the remaining two are temporal-straddle exclusions, one of which is severe. That single severe straddle event accounts for the difference between the eligible cohort (703 severe) and the analysis-assigned set (702 severe), so eligibility is not the same as analysis-role assignment (Table 2).

Endpoint.

We label an event severe when its event-maximum CDI is at least 6.0 (intensity VI or greater). We use 6.0 as the primary threshold and report sensitivity at 5.5, 6.0, and 6.5. We also examine a near-threshold band of ± 0.5 (Section 5.2).

Outcome-boundary disclosure and held-out semantics.

Only corpus-level outcome-blind counts and inherited aggregate feasibility statistics were exposed before construction; no per-event label informed the endpoint, split, or tuning choices, and role membership is assigned by an internally specified, outcome-independent rule that never uses the label. We therefore describe the external holdout as *held out from model development with aggregate feasibility previously known*, and we deliberately do *not* call it “untouched” or “confirmatory.” Two event identifiers were prelisted for quarantine because their aggregate counts had been seen during source verification; neither identifier occurs in the frozen snapshot, so the rule is non-operative in version 1 and excludes no record. The complete list of protocol decisions is reported in Section 5.8. No respondent-level personally identifying information is redistributed; only event-level source metadata, the severe label, the eligibility-only response count, identifiers, and role assignments are released. Snapshot-level provenance is supplied by the activated source-locator manifest and source payload SHA-256; version 1 does not claim per-event product-version hashes. The release bundle follows datasheet-for-datasets documentation [8] and findable, accessible, interoperable, and reusable (FAIR) data principles [36].

4 Benchmark design

4.1 Feature sets and the six-category leakage audit

The primary *source-only* feature set contains magnitude and depth. It is the only feature set evaluated on the temporal holdout. An *expanded-metadata* sensitivity analysis (baseline B3) additionally uses latitude, longitude, and event year. Event year is known at prediction time and is not intrinsically label leakage, but it is excluded by the declared source-only temporal-holdout contract. The audit accepts only an approved feature set with the expected fields in the expected order; any extra or unknown field is rejected.

The executable audit enforces six leakage families. Field-level: F1 direct label or its constituents; F2 post-outcome ascertainment or revision (including the response count, eligibility-only); F3 identifier or source proxy; F4 downstream-product proxy. Split-level: F5 sequence cross-role straddle; F6 temporal-cutoff violation. An axis-variable check additionally rejects origin year under a temporal holdout and, when a geographic holdout is configured, latitude/longitude/region. The latter audit branch is provided for reuse but was not executed as an external evaluation in this study. This enforces a conservative B3 scope restriction; it is not evidence that event year is a leaking label proxy. The audit is useful independently of model performance: it must reject every deliberately contaminated test case and accept only the approved clean feature set. It is exercised by 110 unit tests across four suites, all passing, and accepted the source-only feature set in the frozen snapshot.

4.2 Sequence grouping and outcome-independent roles

Events are grouped into seismic *sequences* by order-stable single-link space-time connected components (100 km, 30 days); the sequence, not the event, is the unit of role assignment, so events linked by this fixed rule cannot straddle roles. This rule is a deterministic grouping heuristic, not a validated catalogue of all physical

mainshock–aftershock relationships; unlinked true aftershocks may therefore remain in different roles. Roles are assigned by an internally specified, outcome-independent rule (fixed assignment key, temporal cutoff year 2023, development fraction 0.2, five cross-validation folds); the rule provably never reads the label. The external holdout contains events on or after 2023. The frozen internal protocol also named a secondary leave-region-out geographic evaluation, but that branch was not executed and no geographic role manifest is claimed. The reported geographic result instead summarizes the temporal-holdout predictions by 10° cell (Supplementary Table S3); Supplementary Table S2 records this deviation. Digital fingerprints of the completed temporal role assignments are published so that third parties can verify them. The partition is exhaustive and disjoint: $1,573 + 357 + 430 + 2 = 2,362$ events, comprising 1,573 train, 357 development, 430 external temporal holdout, and 2 temporal-straddle exclusions (Table 2, Fig. 2). Construction (train plus development) is 1,930 events across 1,530 sequences; the holdout is 430 events across 354 sequences.

4.3 Baseline suite and exclusions

The fixed suite is B0 no-skill (constant prevalence), B1 magnitude-only logistic regression, B2 source-only logistic regression, B3 expanded-metadata logistic regression (used only in compatible sensitivity analyses), B4 random forest (scikit-learn; 21), and B5 XGBoost [4]. Each learned baseline is tuned *only* by the protocol-defined deterministic grouped cross-validation on the construction corpus over a fixed grid, named before evaluation, and carried unchanged to the protected holdout with frozen best parameters. A machine-readable fit log enforces a maximum of 120 model fits; the executed run used 91. Two comparators are excluded for protocol-defined, reproducible reasons recorded in the protocol-decision table (Section 5.8): B3 is not evaluated on the temporal holdout because origin year is outside the declared source-only feature contract; and the Atkinson–Wald (2007) DYFI intensity-prediction equation [2]

234 is excluded because it cannot be applied to the source-only feature set and tectoni-
235 cally heterogeneous population without inventing a site-distance covariate, a per-event
236 tectonic-regime assignment, and a probability link.

237 4.4 Metrics, calibration, uncertainty, and ranking

238 The primary score is the Brier score [3]; secondary scores are log-loss, AUROC,
239 and calibration intercept/slope [31]. For numerical stability, log-loss clips predicted
240 probabilities symmetrically to $[10^{-15}, 1 - 10^{-15}]$ before taking logarithms. Brier-
241 score uncertainty is quantified by a sequence-cluster bootstrap ($n_{\text{boot}} = 2000$, 95%
242 intervals) that resamples whole sequences rather than events [7], with a minimum-
243 valid-replicate rule. Calibration and feasibility states are *derived*, not hard-coded: a
244 constant-prevalence predictor has no identifiable calibration slope, so the table reports
245 that value as unavailable. Pairwise ranking uses a paired sequence-cluster bootstrap
246 Brier difference with an explicit tie probability; the table reports whether each 95%
247 interval includes zero, together with a tie-aware joint ordering (Section 5.6). Because
248 the intervals use 2,000 Monte Carlo draws, near-zero endpoints are interpreted cau-
249 tiously rather than as stable ranking guarantees. The protocol requested intervals for
250 every metric; only Brier-based intervals were retained, so other metrics are descriptive
251 point estimates (Supplementary Table S2).

252 4.5 Protected evaluation and reproducibility

253 The official decision-bearing protected evaluation was authorized once by an auto-
254 mated, globally consumed token; the one-shot wrapper refuses a second authorized
255 run. After the result was fixed, the token-free deterministic analytic core—including
256 holdout scoring—was replayed twice for verification, and both replays reproduced the
257 recorded result fingerprint exactly. Thus, single-use refers to result authorization and
258 decision-bearing use, not to the total number of post-result computations. The frozen
259 holdout predictions were also recovered bit-for-bit from the released rows and frozen

best parameters (reported in Section 5.8), under a pinned scientific-Python environment (NumPy, SciPy, scikit-learn, XGBoost; 4, 9, 21, 32). All fits, exclusions, subsets, cutoffs, and pairwise comparisons are reported regardless of outcome; the only permitted omissions are the protocol-defined exclusions above. Computation is local-only under a pinned environment.

Reproducible artwork and generative AI assistance.

All figures were generated with Matplotlib 3.11.1 directly from the frozen analysis assets; no manual image manipulation or AI-generated artwork was used. ChatGPT assisted language editing, formatting, code review, literature discovery, critical consistency review, and revision suggestions. It did not generate data, execute model fits, or create the reported numerical results. The authors independently verified every cited source, computation, interpretation, and assisted change; all scientific judgments and final decisions remained with the authors, who retain full responsibility for the manuscript and released artifacts.

5 Results

All result-bearing values below are generated from the frozen analysis outputs and checked automatically against their source records; no result cell was hand-edited.

5.1 Cohort and roles

Table 2 and Fig. 2 give the cohort accounting. Of 7,520 frozen DYFI records, 2,362 events pass the ten-response eligibility filter (severe prevalence 0.298; 703 severe events), partitioned into 1,573 train, 357 development, and a 430-event temporal hold-out, with two temporal-straddle exclusions. Thus 2,360 of the 2,362 eligible events receive analysis roles; they contain 702 severe events, and the 703rd severe event is one of the two straddle exclusions. Two identifiers prelisted for quarantine are absent

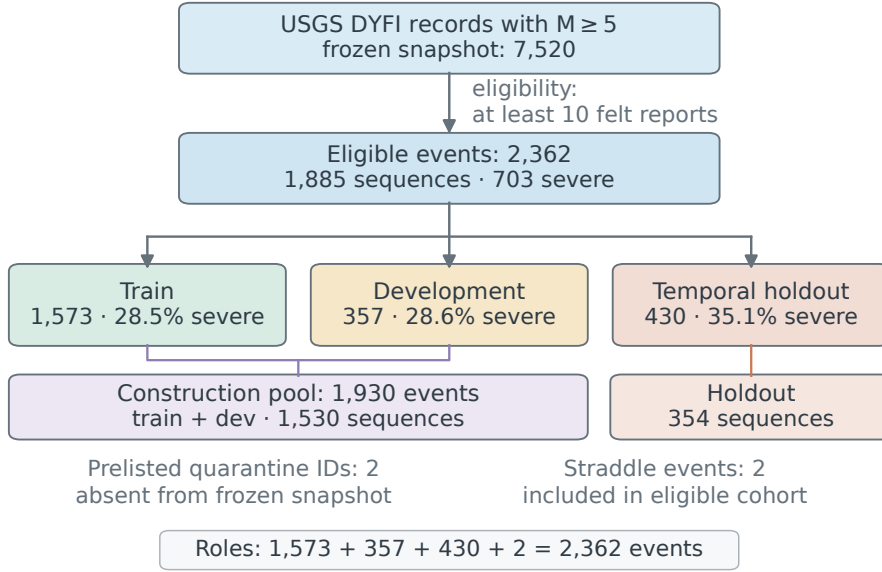


Fig. 2 Cohort construction and role assignment. On the frozen USGS DYFI $M \geq 5$ snapshot. The 7,520 source records yield 2,362 eligible events in 1,885 seismic sequences. Of these, 2,360 are assigned to train ($n = 1,573$), development ($n = 357$), or temporal holdout ($n = 430$); two straddle events are excluded from role assignment

Table 2 Cohort accounting and outcome-independent role assignment

Stage	Events	Severe	Severe rate	Sequences
USGS DYFI records with $M \geq 5$ (frozen snapshot)	7520	–	–	–
Eligible (at least 10 felt reports)	2362	703	0.298	1885
Train	1573	449	0.285	–
Development	357	102	0.286	–
Construction (train+dev)	1930	551	0.285	1530
External temporal holdout	430	151	0.351	354
Temporal-straddle excluded	2	1	–	1

Note. Roles were fixed without using outcomes. Two event identifiers prelisted for quarantine are absent from the frozen snapshot, so the safeguard removes no version 1 record. Severe = 1 when an event's maximum community decimal intensity (CDI) is at least 6. The eligible severe count reconciles as $703 = 449$ (train) + 102 (development) + 151 (holdout) + 1 (straddle exclusion). Thus the 2360 role-assigned events contain 702 severe events.

284 from the frozen snapshot, so that safeguard removes no version 1 record. Construc-
 285 tion severe prevalence is 0.285 and holdout prevalence is 0.351; the higher holdout
 286 prevalence reflects the temporal transport shift and is reported, not adjusted.

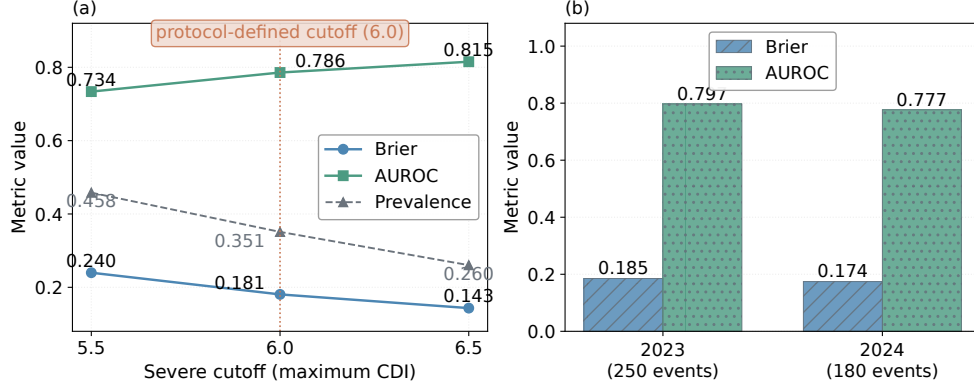


Fig. 3 Endpoint and temporal sensitivity. Results for the source-only logistic regression (B2). (a) Performance as the severe-intensity cutoff varies. (b) Performance within the 2023 ($n = 250$) and 2024 ($n = 180$) portions of the temporal holdout. The cutoff analysis rescores fixed B2 probabilities without refitting. Lower Brier score and higher AUROC are better

Table 3 Endpoint-threshold probe using fixed source-only probabilities

Maximum CDI cutoff	Severe rate	Brier score	AUROC
5.5	0.458	0.240	0.734
6.0 ^a	0.351	0.181	0.786
6.5	0.260	0.143	0.815

Note. ^a Protocol-defined primary cutoff (≥ 6). CDI denotes community decimal intensity. The sensitivity rows relabel and rescore fixed B2 probabilities; B2 is not refit or recalibrated for the alternate cutoffs. In the construction sample, 0.164 of events lie within 0.5 CDI units of the primary cutoff, and the no-skill Brier score is 0.204. Lower Brier score and higher AUROC indicate better performance.

5.2 Difficulty and endpoint sensitivity

Table 3 and the left panel of Fig. 3 report a descriptive fixed-prediction threshold probe. As the severe cut moves from 5.5 to 6.5, holdout prevalence falls smoothly (0.458, 0.351, 0.260) and the source-only logistic-regression Brier score and AUROC move monotonically (Brier 0.240, 0.181, 0.143; AUROC 0.734, 0.786, 0.815). On the construction corpus the near-threshold fraction ($|\text{maxCDI} - 6| < 0.5$) is 0.164 and the no-skill Brier score is 0.204, bounding the difficulty of the primary endpoint. This probe relabels and rescores probabilities fitted for the primary 6.0 endpoint; it does not refit or recalibrate B2 at 5.5 or 6.5 and therefore does not establish endpoint robustness. The protocol-required aggregation-grid sensitivity was not executed and is disclosed as a deviation (Supplementary Table S2).

5.3 Observed split-performance differences

The resource’s headline quantity is the optimism of naive random splitting relative to the sequence-grouped protocol, measured for a fixed leakage diagnostic—a random forest with 300 trees and unrestricted depth—under three fold-assignment schemes on the construction corpus (Table 4, Fig. 4). This diagnostic is distinct from the tuned temporal-holdout baseline B4, whose maximum depth is 3. On this $M \geq 5$ one-row-per-event universe the point estimates are small and slightly reversed: sequence grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than random-event splitting, calculated from unrounded scores. The sequence-grouped split therefore scores marginally *better* than the naive random split, not worse. The frozen internal protocol required uncertainty reporting for this contrast but did not define a paired interval estimator; the executed result contains point estimates only. We disclose this unfulfilled element as a deviation and do not invent a post-hoc interval. Accordingly, we report the contrast descriptively, not as evidence of equivalence, a true zero effect, leakage, or a win. Its plausible structural context is that 1,885 sequences are largely singleton (1,530 in construction), leaving limited within-sequence near-duplicate structure for a random split to leak. This does not prove a small true effect. With one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation differs and therefore produces different scores. It is reported as an intermediate control. The protocol remains reusable on datasets with more multi-event sequences. The present result neither demonstrates practically important leakage from random splitting nor establishes its absence.

5.4 Baseline performance and calibration

Table 5 and Fig. 5 give per-baseline holdout performance ($n = 430$) with sequence-cluster bootstrap 95% intervals. The no-skill B0 has Brier 0.232 ([0.212, 0.253]); its

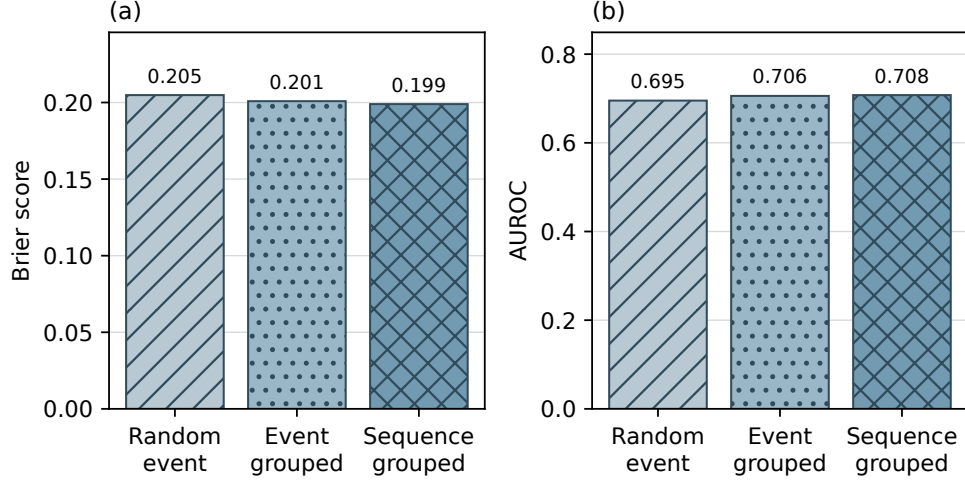


Fig. 4 Split-assignment diagnostic. Five-fold performance of the fixed diagnostic (random forest: 300 trees, unrestricted depth) under random-event, event-grouped, and sequence-grouped fold assignment in the construction sample. Relative to random-event splitting, sequence grouping produced (a) a Brier score 0.0059 lower and (b) an AUROC 0.0124 higher. The diagnostic is distinct from tuned temporal-holdout baseline B4. The small, reversed differences provide no evidence of performance inflation from random splitting in this sample. Both axes start at zero

Table 4 Split-assignment performance of the fixed diagnostic

Fold assignment	Brier score	AUROC	Log-loss	Folds
Random-event (naive)	0.2048	0.6952	0.9640	5
Event-grouped (control)	0.2009	0.7058	0.9223	5
Sequence-grouped (primary)	0.1990	0.7076	0.9709	5

Note. The diagnostic is distinct from the tuned temporal-holdout baseline B4 (depth 3). Using unrounded values, sequence grouping produced a Brier score 0.0059 lower and an AUROC 0.0124 higher than random-event splitting. The frozen internal protocol required uncertainty but did not define the paired interval estimator; the executed report retained point estimates only. This deviation is disclosed, and no post-hoc interval or equivalence claim is added. Because there is one row per event, event grouping imposes the same grouping constraint as random-event assignment, but its deterministic fold allocation is different and can produce different scores.

324 calibration slope cannot be estimated because it assigns the same probability to every
325 event. The learned baselines have lower Brier (B1 0.198, B2 0.181, B4 0.187, B5 0.181)
326 with overlapping intervals; calibration intercept/slope pairs are finite and positive. No
327 baseline is claimed to win; the table reports proper scores, uncertainty, and derived
328 calibration states only.

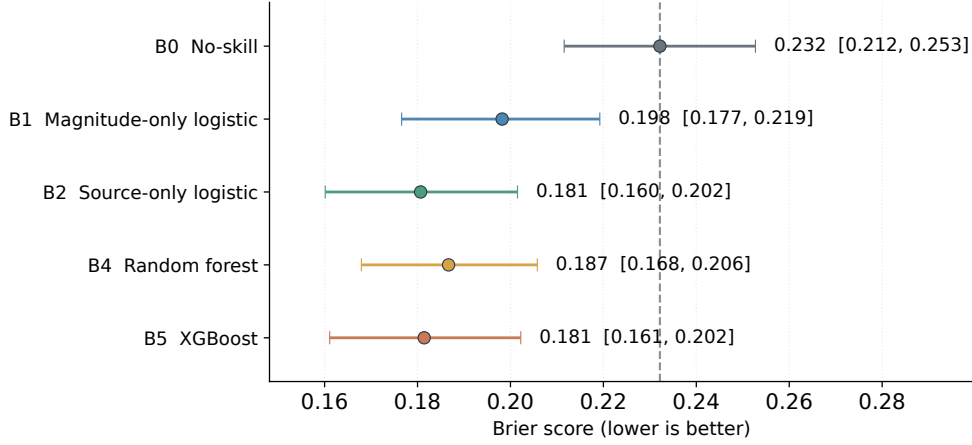


Fig. 5 Temporal-holdout baseline performance. Brier scores and 95% sequence-cluster bootstrap confidence intervals for the five baselines on the temporal holdout ($n = 430$). Lower values are better; the vertical dashed line marks the no-skill B0 score. B3 is omitted because event year is outside the declared source-only temporal-holdout contract (Supplementary Table S2)

Table 5 Temporal-holdout performance of fixed baseline models

Baseline	Brier	95% CI	Log-loss	AUROC	Calibration ^a	Accuracy
B0 (no-skill)	0.232	[0.212, 0.253]	0.658	0.500	INFEASIBLE	0.65
B1 (magnitude-only logistic)	0.198	[0.177, 0.219]	0.583	0.719	0.32 / 1.02	0.70
B2 (source-only logistic)	0.181	[0.160, 0.202]	0.536	0.786	0.64 / 1.29	0.73
B4 (random forest)	0.187	[0.168, 0.206]	0.553	0.777	0.74 / 1.48	0.73
B5 (XGBoost)	0.181	[0.161, 0.202]	0.541	0.783	0.42 / 1.07	0.74

Note. ^a Intercept / slope. These values cannot be estimated for B0 because it assigns the same probability to every event. B3 includes event year and is therefore omitted from the temporal holdout (Supplementary Table S2). Accuracy uses a fixed probability cutoff of 0.5 that was not selected on the holdout. Log-loss clips probabilities to $[10^{-15}, 1 - 10^{-15}]$. Lower Brier score and log-loss are better; higher AUROC and accuracy are better. The two lowest-Brier models cannot be distinguished in their direct paired comparison.

5.5 Worked candidate qualification

Table 6 applies the released engineering decision rule to the six fixed candidates. B3 is outside the source-only contract because it contains event year; this is a conservative scope exclusion, not demonstrated label leakage. B1, B2, B4, and B5 pass the audit and each has a paired Brier interval below the no-skill B0 reference. The contract therefore supports their improvement over B0. It does not support naming B2 or B5 as the winner: their direct paired interval includes zero. This worked example

Table 6 Worked candidate-model qualification using the frozen temporal holdout

Candidate	Gate	Holdout evidence	Decision
B0 no-skill	Pass	Brier 0.232 [0.212, 0.253]; declared reference	Reference
B1 magnitude-only logistic	Pass	Brier 0.198 [0.177, 0.219]; Δ vs B0 -0.0339 [-0.0511, -0.0171]	Supported
B2 source-only logistic	Pass	Brier 0.181 [0.160, 0.202]; Δ vs B0 -0.0515 [-0.0681, -0.0342]	Supported
B4 random forest	Pass	Brier 0.187 [0.168, 0.206]; Δ vs B0 -0.0455 [-0.0598, -0.0307]	Supported
B5 XGBoost	Pass	Brier 0.181 [0.161, 0.202]; Δ vs B0 -0.0507 [-0.0682, -0.0319]	Supported
B3 expanded-metadata logistic	Out of contract	—	Year excluded by schema

Decision rule. A candidate is first admitted by the schema, proxy, role, and temporal audits. A lower paired Brier interval that excludes zero supports improvement over the declared reference; otherwise the comparison remains unresolved. All four learned baselines improve over B0. B2 versus B5 remains unresolved (B2–B5 -0.0008, 95% CI [-0.0078, 0.0068]), so the contract prevents a false winner declaration.

demonstrates the benchmark’s intended use—admit only valid candidates, support improvements with uncertainty, and leave an unresolved comparison unresolved.

5.6 Pairwise ranking stability

Supplementary Table S1 reports all ten pairwise Brier-score comparisons. Every learned baseline performs better than the no-skill reference B0 (all four intervals exclude zero). Among the learned models, the B2–B4 and B2–B5 comparisons have intervals that include zero. The remaining learned-model intervals exclude zero, although the B4–B5 lower endpoint (0.0003) is near the Monte Carlo boundary. In the joint bootstrap ordering, B2 has the highest probability of being uniquely best (0.613), followed by B5 (0.387), but the direct B2–B5 comparison remains uncertain. These results therefore do not justify naming a winning model.

The frozen internal protocol allowed an “equivalent” verdict but did not specify a positive numerical equivalence margin; its resolution floor was zero. We therefore do not invent a post-hoc margin or label any models equivalent. Instead, we report each point difference, 95% interval, and whether that interval includes zero. Supplementary Table S2 records this interpretive restriction, which changes no result.

352 5.7 Temporal and geographic subgroup summaries

353 Supplementary Table S3 and the right panel of Fig. 3 report the temporal and
354 geographic subgroup summaries for the source-only logistic regression within the tem-
355 poral holdout. The two holdout years are internally consistent (2023: $n = 250$, Brier
356 0.185, AUROC 0.797; 2024: $n = 180$, Brier 0.174, AUROC 0.777). The geographic
357 subgroup summary spans 82 occupied 10° cells (per-cell Brier 0.002–0.825, median
358 0.151). These values stratify the same 430 temporal-holdout predictions; they are not a
359 leave-region-out evaluation and support no geographic or cross-system generalization
360 claim.

361 5.8 Exclusions, deviations, and reproducibility

362 Supplementary Table S2 explains the protocol decisions most relevant to inter-
363 pretation: excluded comparators and records, the unexecuted geographic transport
364 branch, the unfulfilled split-contrast interval, and the absence of an equivalence
365 claim. The tagged public release contains the complete SHA-256 fingerprints, software
366 environment, and machine-readable verification details. Its public-release manifest
367 fingerprints every distributed file. Three original execution-bookkeeping records are
368 self-referential—they cannot contain their own final fingerprints—and are therefore
369 reported separately rather than treated as analysis-file mismatches. A machine-
370 readable corrections overlay deprecates three legacy semantic fields (protected-role
371 status, the generic split-interpretation template, and event-group wording) and
372 provides authoritative replacements without changing any numerical result.

373 6 Discussion

374 *Engineering use and decision consequence.*

375 The resource supports a transparent retrospective audit of an event-level classifier.
376 The user does not choose a convenient split or reconstruct a moving dataset; the

377 declared candidate is evaluated against a released contract. The audit rejects forbid-
378 den outcome-derived inputs, the role manifest prevents events linked by the declared
379 heuristic from crossing roles, and the paired Brier interval determines whether improve-
380 ment over a reference is supported. The worked example changes two decisions: B3 is
381 outside the source-only contract, while B2 and B5 are both supported over no-skill
382 but cannot be ranked decisively. Without the contract, the same point scores invite
383 an unsupported winner claim. Because all labels are public, a developer can tune to
384 this release; prospective or confirmatory qualification requires a fresh hidden holdout
385 or independent evaluation service.

386 *Interpreting the split contrast.*

387 The observed point estimate is small and slightly reversed. In these folds the naive
388 random split did not inflate the diagnostic score, but the missing interval pre-
389 vents an equivalence, zero-effect, or “little practical effect” inference. The protocol
390 remains useful for datasets containing more multi-event sequences; this result neither
391 demonstrates leakage nor supports a win.

392 *No winning model.*

393 Every learned baseline separates from the no-skill reference. However, the direct B2–
394 B5 comparison between the two lowest-Brier models is unresolved, as is B2–B4, so
395 the results do not establish a decisive winner. Consistent with the resource’s scope,
396 we claim no winning model and no operational, real-time, lead-time, cross-system, or
397 causal benefit.

398 **7 Limitations and threats to validity**

399 *Outcome and catalogue limitations.*

400 The endpoint is magnitude-dominated, CDI remains sensitive to responder coverage,
401 and magnitude and depth are frozen preferred catalogue values rather than guaranteed

402 first-alert values. The ten-response rule sets an eligibility and reliability floor but
403 selects on post-event participation and does not remove ascertainment; response count
404 is used only for eligibility. B1 provides the simple magnitude comparator.

405 *Unexecuted sensitivity and transport branches.*

406 The geographic values are temporal-holdout subgroups, not external transport evi-
407 dence. The protocol-defined geographic evaluation and aggregation-grid sensitivity
408 were not executed, and no alternative to the fixed 100-km/30-day sequence defini-
409 tion was tested. The rule guarantees co-assignment only for events it links; it does
410 not guarantee identification of every physical aftershock relationship. Consequently,
411 no geographic, aggregation, sequence-definition, or complete aftershock-isolation
412 robustness claim is made.

413 *Inference and held-out semantics.*

414 The protocol-required split-contrast uncertainty interval was not retained, so that
415 result remains descriptive with no equivalence or zero-effect inference. Non-Brier
416 scores remain descriptive point estimates. Pairwise intervals use 2,000 Monte Carlo
417 draws and near-zero endpoints are interpreted cautiously. The temporal holdout was
418 excluded from model development, but aggregate feasibility was previously known; it
419 is neither “untouched” nor confirmatory.

420 *Public release.*

421 Version 1 of the data, code, frozen results, and reproduction instructions
422 is available in a tagged public repository at [https://github.com/akssha74/](https://github.com/akssha74/dyfi-benchmark-reliability)
423 [dyfi-benchmark-reliability](https://github.com/akssha74/dyfi-benchmark-reliability). The release retains complete digital fingerprints so that
424 later revisions remain distinguishable from the version evaluated here.

425 8 Conclusion

426 Reliable engineering model selection requires more than reporting the lowest point
427 score. The released DYFI benchmark makes a candidate pass an auditable schema
428 and role contract, evaluates it on a fixed temporal holdout, and attaches paired uncer-
429 tainty to the comparison decision. In the worked qualification, four learned candidates
430 improve over no-skill, B3 is outside the declared source-only contract, and the two
431 lowest-Brier models remain unresolved; the contract therefore prevents an out-of-scope
432 comparison and a false winner declaration. The random-versus-sequence diagnostic is
433 small and reversed in this sparse-sequence corpus, so it supplies no evidence of random-
434 split optimism. All results reconstruct from the frozen release. The contribution is a
435 reusable retrospective test harness for trustworthy model comparison. Confirmatory
436 use requires a fresh hidden holdout or independent evaluation service; the release is
437 not a real-time warning, causal model, or operational decision system.

438 Declarations

- 439 • **Funding.** This research did not receive any specific grant from funding agencies in
440 the public, commercial, or not-for-profit sectors.
- 441 • **Competing interests.** The authors have no relevant financial or non-financial
442 interests to disclose.
- 443 • **Ethics approval.** Not applicable. The study analyzes public, aggregate event-level
444 USGS products rather than individual questionnaire responses; it involved no inter-
445 action with participants, identifiable data, human-subject collection, animals, or
446 biological material.
- 447 • **Consent to participate.** Not applicable; no participants were recruited.
- 448 • **Consent for publication.** Not applicable.

- 449 • **Availability of data and materials.** Version 1 data, results, manifests,
 450 and instructions are available under CC0 1.0 at [https://github.com/akssha74/](https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.1.1)
 451 [dyfi-benchmark-reliability/releases/tag/v1.1.1](https://github.com/akssha74/dyfi-benchmark-reliability/releases/tag/v1.1.1). Source data are USGS DYFI and
 452 ANSS ComCat (public domain; [10.5066/F7J101C8](https://doi.org/10.5066/F7J101C8) and [10.5066/F7MS3QZH](https://doi.org/10.5066/F7MS3QZH)) and
 453 are reconstructable via the released fetch-and-verify source-locator manifest. No
 454 respondent-level personally identifying information is redistributed.
- 455 • **Availability of code.** Project: DYFI Benchmark Reliability. Homepage: [https://](https://github.com/akssha74/dyfi-benchmark-reliability)
 456 github.com/akssha74/dyfi-benchmark-reliability. Archived version: release v1.1.1.
 457 Operating systems: platform independent. Programming language: Python. Licence:
 458 MIT. Restrictions on non-academic use: none. Full prediction reconstruction and
 459 model tests use the separately pinned scientific-Python environment.
- 460 • **Author contributions.** A.S. conceived and designed the benchmark protocol,
 461 developed the software, curated the data, performed the analyses, validated the
 462 results, and drafted the manuscript. L.P. contributed to study design and inter-
 463 pretation, supervised the work, and critically revised the manuscript for important
 464 intellectual content. Both authors reviewed and approved the final manuscript and
 465 agree to be accountable for all aspects of the work.
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